

"Indian Institute of Technology, Madras"

Department

of

Mechanical Engineering

✦ Experiment No. : 04 ✦

"Programming of Microprocessor"

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Experiment No. - 4

Programming of Microprocessor

* Aim: → To learn assembly level language of coding and execution of program in 8085 microprocessor based kit

* Evolution of Processor: →

- ① Main Branch: Comprises processor developments to realize general purpose computers starting from 4 bit 4004 processor to 64 bit processor having these important features:
 - Pipelining
 - Superscalar processor
 - Cache (Pentium, 4EE, 4E, 4F, Xeon)
- ② Embedded System (Microcontroller): All in one system, no input output device to interface, so it is a complete system to do specific work eg. TV remote, washing machine, ATM machine (Intel - 8048, 80251)
- ③ Special Purpose Chip: Teleconferencing; communicate (cell phone) TMS-320.

* Equipments : → 8085 Microprocessor kit

1. 8085 CPU of 6.144 MHz clock speed.
2. RAM - 8KB expandable to 32 kb, jumper selectable.
3. EPROM - 8KB expandable to 32 kb, jumper selectable.
4. NiCd Battery (or) Lithium battery backup for RAM.

• Peripherals : "keyboard Display"

1. PC keyboard 104 keys.
2. 16x2 Alphanumeric dot matrix liquid crystal display with backlight timer parallel input.

* Procedure for making codes / Assembly language in

8085 kit : →

1. Plug in power supply and keyboard.
2. Switch on the power.
3. Type 4 in the processor.
4. Default address is 4200 if we want to change the address to 4100 type ORG <space> <required address>
5. Enter all memories in the address.
6. When the program ends (HLT) press enter and reset.
7. To input values, type SU and type the input address in which the value should be entered.
8. Press reset.

* Procedure to execute the Programme:->

1. To execute the program type G < space > < starting address > and enter.
2. To check the output type SU < space > < output address / accumulator > and enter.
3. Observe the output result from the display.

* Programme:->

Add two 8 bit numbers stored in the memory location 4150 and 4151 and store the result in 4152 memory location. If the result has carry then store the carry to 4153 memory location.

* Algorithm:->

Memory address	Mnemonics	Opcode
4201	MVIC, 00	0E
4203	LDA 4150	3A
4206	MOV B, A	47
4207	LDA 4151	3A
420A	ADD B	88
420B	JNC (loop)	82
420E	INR C	0C
420F	LOOP: STA 4152	32
4212	MOV A, C	79
4213	STA 4153	32
4214	HLT	76

* Keyboard Inputs with explanation :-

SU 4201 - Go to the memory location 4201.

4201 0E }
4202 00 } Load reg. C with the value 00

4203 3A }
4204 50 } Load the accumulator with the contents from
4205 41 } memory address 4150

4206 47 - Move the value from acc. to reg. B

4207 3A }
4208 51 } Load the accumulator with the contents from
4209 41 } memory address 4151.

420A 88 - Add contents of acc. with contents of reg. B

420B D2 }
420C 01 } Jump to the memory address 420F if there is
420D 42 } no carry.

420E 0C - Increase the value stored in reg. C by 1.

420F 32 }
4210 52 } store accumulator contents in the memory
4211 41 } address 4152

4212 79 - move the value of reg. C to acc. (A).

4213 32 }
4214 53 } store accumulator contents in the memory
4215 41 } address 4153

4216 76 - Halt the microprocessor (Ends the programme).

* Observation : $\rightarrow$

1. I/P : 2F (4150)
6D (4151)

O/P : 9C (4152)
00 (4153) $\rightarrow$ No carry

2. I/P : 9F (4150)
8E (4151)

O/P : 2D (4152)
01 (4153) $\rightarrow$ carry